



## Forensic aspects of homicides by insulin overdose

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### ABSTRACT

Analysis of homicidal insulin overdose is a challenging task in forensic practice because of the difficulties in toxicological analysis as well as the elusive pathologic changes. We performed a detailed histopathologic examination on four autopsy cases involving insulin homicide, using H&E, immunohistochemistry (IHC) and immunofluorescence assays. Severe reactive astrocyte proliferation was obvious in the white matter of the cerebrum, corpus callosum, cerebellum and brain stem, especially in subcortical regions. We found a statistically significant increase in the number and total area of reactive astrocytes compared with controls ( $p < 0.001$ ). Insulin was detected at the injection sites of subcutaneous soft tissues by using IHC, luminescence immunoassay and immunofluorescence. Most insulin deposits were located in the gaps between adipocytes, and a few deposits were located in peripheral nerves and inflammatory cells. We also detected protamine in the skin tissues in two of the four cases. Our study revealed that the presence of insulin and/or protamine at the injection sites, along with severe reactive astrocyte proliferation, could help diagnose insulin overdose.

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### 1. Introduction

Insulin is a hormone which regulates glucose concentration. It consists of two peptide chains (A and B chains) which are connected by a disulfide bond. The molecular weight of insulin is 5807.69. The plasma half-life of insulin is short, approximately 4–6 min [1–4]. Proinsulin molecules also consist of two chains along with a connecting peptide (C-peptide) before secretion. Equimolar amounts of C-peptide and insulin will be released into circulation when insulin is secreted following hyperglycemia [5–8]. The plasma half-life of C-peptide is about 10–20 min [4]. Exogenous insulin administration can suppress the secretion of C-peptide due to a negative feedback system [9]. Subsequently, the C-peptide/insulin ratio in blood will decrease. According to Uezono et al., analysis of serum C-peptide concentration is necessary to diagnose hypoglycemia induced by exogenous insulin [10]. Cases of insulin

overdose mostly occur in clinical practice and fatal cases are rarely reported [11–16].

In modern clinical practice, insulin is widely used to treat diabetes, but its administration should be carefully controlled since it could give rise to adverse results, such as severe hypoglycemia, coma and death. In the 1930s, insulin coma therapy (ICT) was thought to be a revolutionary treatment for schizophrenia [17], but in the 1950s, ICT was questioned [18,19] and gradually abolished in treating mental diseases.

Insulin overdose could be exploited as a murder weapon [17]. The first insulin homicide was reported in 1958 by Birkinshaw [20]. Some fictional writers have described insulin as the perfect murder weapon because of the elusive pathologic changes and the difficulties in toxicological screening [12]. In cases of fatal insulin overdose, accidental overdose is the most common manner of death, followed by suicide, whereas homicide is extremely rare [6,21]. In recent times, several cases of homicide due to insulin overdose have been reported in mainland China and such cases now tend to appear more frequently [22–25].

The purpose of this study is to investigate the pathologic changes resulting from insulin overdose in a series of established homicidal insulin overdose cases using H&E, immunohistochemistry (IHC), and immunofluorescence assays.

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## 2. Materials and methods

### 2.1. Victim information and preparation of human samples

The samples from four victims murdered by insulin overdose and used in this study were collected at Tongji Medical College, Huazhong University of Science and Technology (HUST). Information regarding the four cases is listed in Tables 1–3. The negative controls were chosen from three cases of sudden natural death independent of insulin. Tissue samples of the brain, heart, liver, kidney and skin at the injection sites were fixed in 10% buffered formalin solution for a week. All tissue blocks were dehydrated and embedded in paraffin. Serial sections of 4  $\mu$ m thickness were used for H&E, IHC and immunofluorescence assays. In the control group, the tissues were sampled and tested in the same way. All experimental protocols involving human samples were approved by the victims' relatives and the Human Ethical Commission at Tongji Medical College, HUST.

### 2.2. Histopathology and immunohistochemistry

Sections of the tissues from the brain, heart, liver, kidney and skin at the injection sites were routinely stained with H&E. After

deparaffinization, sections were dipped into 3% hydrogen peroxide solution for 10 min to quench endogenous peroxidase. Microwave-induced epitope retrieval was performed in citrate buffer (pH 6.0) at 100 °C for 30 min (skin specimens for 5 min). After incubation for 30 min with 5% bovine serum albumin to block nonspecific binding, the sections were incubated with insulin primary antibody (1:600, Boster, China), protamine primary antibody (1:2000, Boster, China) or GFAP primary antibody (1:5000, Abcam, UK) in phosphate-buffered saline at 4 °C overnight. Sections were incubated for 50 min at 37 °C with biotinylated secondary antibody and horseradish peroxidase-conjugated streptavidin (Biogood, China), respectively. Staining was visualized with 3, 3'-diaminobenzidine chromogen, and counterstained with hematoxylin. Finally, sections were dehydrated through increasing concentrations of ethanol and xylene. Negative control sections were incubated with non-immune serum of the species from which the primary antibodies were derived. All the sections were examined under a Nikon Eclipse 90i microscope (Tokyo, Japan). A semi-quantitative evaluation of the reactive astrocytes was made by two different investigators without prior knowledge of the IHC results. The degree of pathologic change was graded as follows: (–) none; (+) mild; (++) moderate; (+++) severe. All measurements were obtained using the same magnification of

**Table 1**  
Basic information of the four cases in the study.

| Case no. | Age/gender | Positions of injection sites    | Total injection volume | Perpetrators                    | Survival interval | Brief presentation                                                                                                                                                                                                                                                                                                                                                       |
|----------|------------|---------------------------------|------------------------|---------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 27 Male    | Left upper arm, left opisthenar | 800 IU/10 ml           | Lover and ex-girlfriend (nurse) | <2 h              | The woman and the victim were having an affair, and she wanted to end the relationship. One evening, she injected 200 IU insulin into the man's left upper arm, but lied that it was vaccine. After he lost consciousness, she injected the remainder of the insulin intravenously at the left wrist. She then dumped the body into a lake after confirming he was dead. |
| 2        | 27 Male    | Right abdomen, left elbow       | 400 IU/10 ml           | Girlfriend (nurse)              | <10 h             | The woman and the victim were going to marry in several months, but marriage plans were cancelled after some trials and tribulations, leading her to take revenge on the male victim. The woman deceived him into drinking a cup of water laced with clonazepam. Once he was asleep, she injected him with insulin.                                                      |
| 3        | 55 Male    | Left arm, middle abdomen        | >1500 IU               | Ex-wife (housewife)             | <8 h              | A woman and the victim had divorced because the man had had an affair, but they still cohabited and had some disputes over property. The woman was an IDDM patient with access to insulin, and after deceiving the victim into taking a drink laced with 2 pills of clonazepam, she injected him with insulin.                                                           |
| 4        | 86 Female  | Abdomen                         | 1200 IU/20 ml          | Daughter (housewife)            | <6 h              | The woman and her mother, the victim, were having difficulties. The mother was an IDDM patient and her daughter regularly injected insulin for her. With an intent to finish a painful life, she injected an overdose of insulin into her mother. She then attempted suicide but failed.                                                                                 |

**Table 2**  
Information about the autopsy and scene investigation.

| Case no. | Postmortem interval before autopsy                   | Scene investigation                                                    | Type of insulin                                                           | Degree of decomposition |
|----------|------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------|
| 1        | 15 days (in water for 3 days and frozen for 12 days) | 2 empty insulin bottles found in the victim's home, the primary scene. | Gansulin (Mixed Protamine Zinc Recombinant Human Insulin Injection)       | Moderate                |
| 2        | About 12 h                                           | 2 empty insulin bottles were found                                     | Humulin (Recombinant Human Insulin Injection)                             | No                      |
| 3        | Frozen for 4 days                                    | >10 empty insulin bottles were found                                   | Novolin (Recombinant Human Insulin Injection)                             | No                      |
| 4        | Frozen for 10 days                                   | 4 empty insulin bottles were found                                     | Insulin Aspart (Mixed Protamine Zinc Recombinant Human Insulin Injection) | No                      |

**Table 3**  
Results of postmortem biochemistry and detection of clonazepam.

| Case no. | Glucose concentration of blood (mmol/L) <sup>a</sup> | Glucose concentration of vitreous humor (mmol/L) <sup>b</sup> | Insulin concentration of blood (mU/L) <sup>c</sup> | Clonazepam of blood |
|----------|------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------|---------------------|
| 1        | 12.16                                                | /                                                             | 2.14                                               | (–)                 |
| 2        | 9.08                                                 | 0.85                                                          | 0.2                                                | (+)                 |
| 3        | /                                                    | /                                                             | 0.2                                                | (+)                 |
| 4        | 11.4 mmol/L                                          | 0.9                                                           | 2.5                                                | (+)                 |

<sup>a</sup> Reference values: 3.9–6.1 mmol/L.

<sup>b</sup> Reference values [9]: approximate 0.2 mmol/L.

<sup>c</sup> Reference values: approximate 5.0–18.0 mU/L.

image (40×, 100×, 200× and 400×) and by the same two examiners. A third blind microscopic evaluator was involved to weigh the histologic changes. Digital pictures of GFAP IHC staining were further measured using Image Pro Plus 6.0 to count the number and sum areas of the reactive astrocytes.

### 2.3. Luminescence immunoassay

One gram of skin tissue from the victims of Case 2, 3, and 4 was ground into 10 ml PBS and centrifuged for 5 min using a high-speed freezing centrifuge (Eppendorf 5810R, Made in Germany) at 4 °C and 12,000 RPM. The supernatant was collected and used to detect insulin using a chemiluminescence immunoassay (BAER ACS-180, Made in Germany).

### 2.4. Double immunofluorescence staining

After deparaffinization, antigen retrieval, and blockage as described above, the sections were incubated with both primary antibodies for mouse-anti insulin (1:800, Boster, China) and rabbit-anti protamine (1:800, Protein Tech, China) overnight at 4 °C, followed by incubation with a mixture of Alexa 488—conjugated anti-rabbit IgG and Alexa 647—conjugated anti-mouse IgG (both 1:1000, Abcam, Carlsbad, USA) for 1 h at 37 °C. Control sections were incubated with non-immune serum instead of primary antibodies. All the sections were examined under a Nikon Eclipse i-S microscope (Tokyo, Japan).

### 2.5. Statistical methods

Data were analyzed using IBM SPSS Statistics 20.0. The threshold for statistical significance was set to  $P < 0.01$  without correction for multiple comparisons. A Wilcoxon rank sum test was used for comparisons of immunostaining profiles between the different groups.

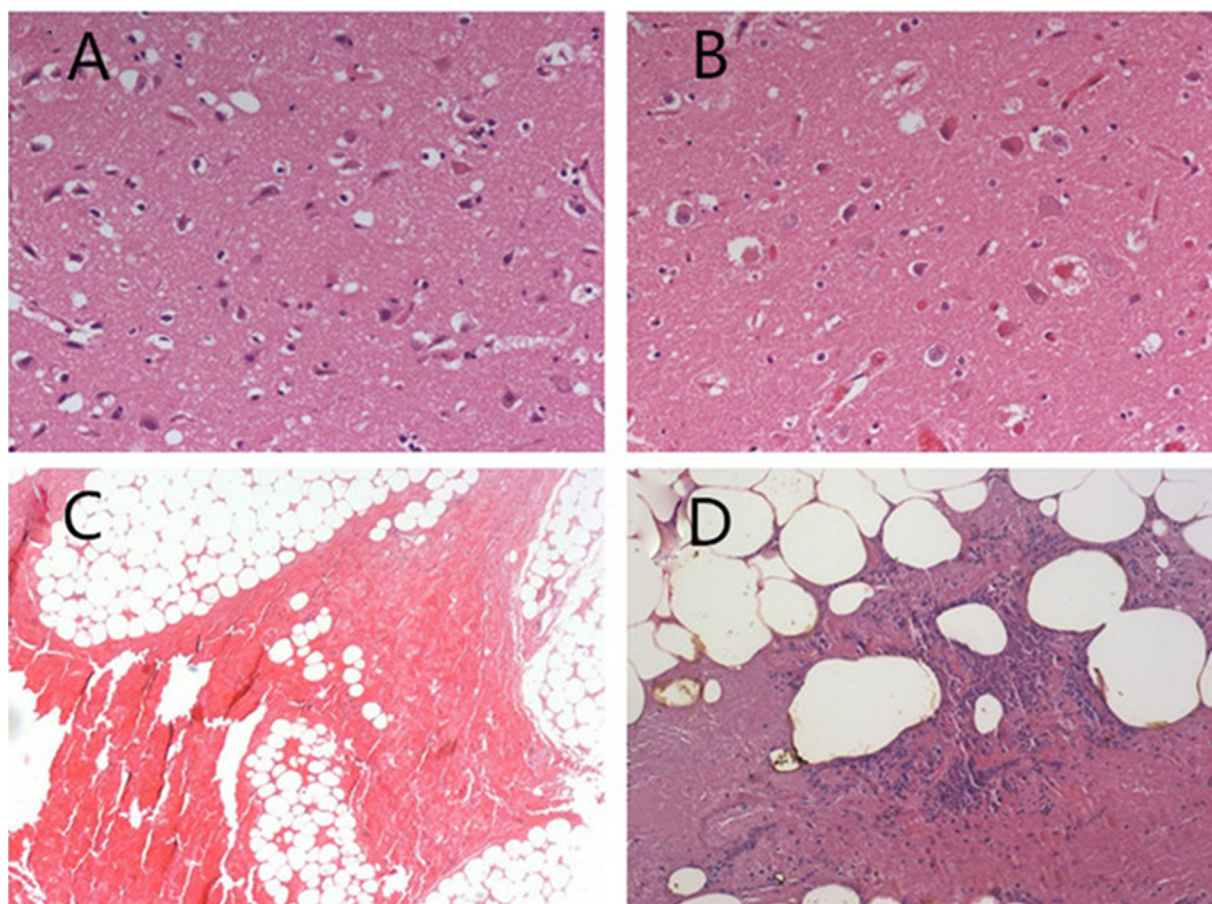
## 3. Results

### 3.1. Histologic findings

**Brain:** Neuronal degeneration/necrosis was conspicuous in the cerebral cortex and basal ganglia (Fig. 1A). Many neurons became pyknotic and red in appearance. Loss of nuclei was observed. Neuronal damage in the cerebellum and brain stem was minor. The parenchyma showed moderate edema and some regions were spongy. Astrocyte proliferation was obvious in the white matter of the cerebrum, corpus callosum, cerebellum and brain stem, especially in subcortical regions. The changes above were observed in all four cases.

**Heart:** No specific pathologic changes were observed except for minor hemorrhages. Occasional inflammatory cells were found in the mesenchyme of Case 4.

**Lung:** Focal edema was observed in the alveolar space. Severe hemorrhage was present in many alveoli. Moderate vascular congestion was also observed.



**Fig. 1.** Histopathology findings in CNS and skin. A: Spongy region and neuronal necrosis/degeneration in the cerebral cortex, H&E 1:200. B: Neuronal necrosis/degeneration in the basal ganglia, H&E 1:200. C: Injection sites of skin, H&E 1:100.



**Table 4**  
Histopathologic changes of the 4 cases.

| Case no. | Necrotic neurons | Necrotic renal tubule epithelium | Renal granular casts | Edema hepatocytes | Necrotic liver tissue | Pulmonary hemorrhage |
|----------|------------------|----------------------------------|----------------------|-------------------|-----------------------|----------------------|
| 1        | +                | +                                | —                    | ++                | ±                     | ++                   |
| 2        | +                | ++                               | +                    | ++                | ±                     | ++                   |
| 3        | +                | ++                               | +                    | ++                | +                     | +                    |
| 4        | +                | ++                               | +                    | ++                | +                     | +                    |

Liver: Hepatocytes showed ballooning and severe edema. Focal necrosis of liver tissues with infiltration of inflammatory cells was observed, which was minor in Case 1 and Case 2, and severe in Case 3 and 4.

Kidney: Epithelial necrosis was observed in both the proximal and distal renal tubules in all four cases. In Case 2, Case 3 and Case 4, the epithelial necrosis was severe. The cytoplasm of some renal epithelial cells was clearly stained red in Case 3 and Case 4. Some of the epithelial cells were fragmented and found in the tubular cavities. Some protein casts were observed in some tubular cavities except for Case 1.

Skin from the injection sites: Various degrees of hemorrhage were observed (Fig. 1C). Some pink-stained liquid in the gaps between the adipocytes (Fig. 1D) was confirmed to be insulin by IHC. Some inflammatory cells were observed in skin tissues. These changes were observed in Case 1, Case 2 and Case 4.

The results of histologic findings are listed in Table 4.

### 3.2. Reactive astrocyte proliferation in the brain

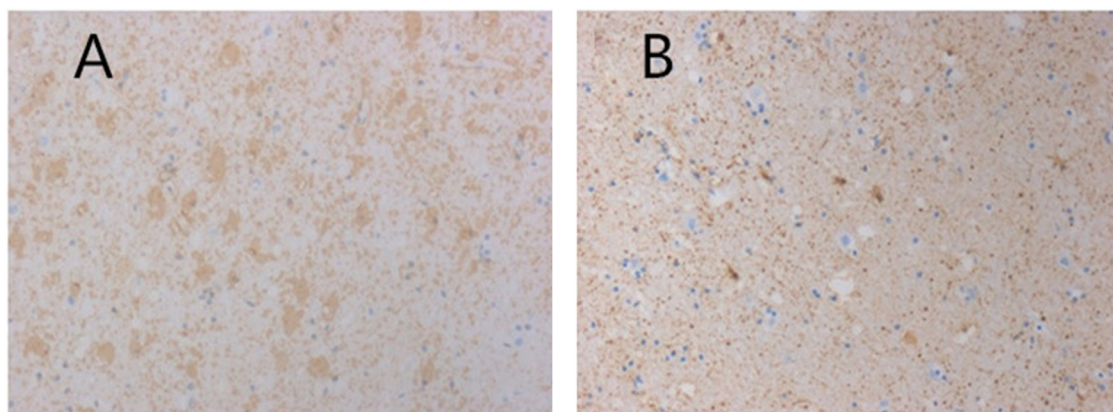
GFAP IHC staining was used to measure the degree of proliferation of reactive astrocytes. Many GFAP IHC-stained reactive astrocytes were observed in the brain of the four victims cases (Fig. 2A) whereas only a few GFAP IHC-stained reactive astrocytes were found in the control cases (Fig. 2B). The results are listed in Table 5. Statistical analysis indicated a significant increase in both the number and total area of reactive astrocytes when compared to controls with 200× field of microscopy ( $p < 0.001$ ) (Fig. 3A and B).

### 3.3. Insulin in injection sites

Insulin was detected in the tissue homogenate of the skin tissues at the injection sites using luminescence immunoassay in Case 2, Case 3 and Case 4. Strongly positive insulin IHC staining areas were observed in the tissues collected from the injection sites (Fig. 4A) in Case 1, Case 2 and Case 4. The accumulation of insulin was found in the gaps between adipocytes (Fig. 4B), which was observed as pink-stained liquid by H&E. Insulin deposits were observed in some peripheral nerves (Fig. 4C). Some inflammatory cells were positive for insulin IHC (Fig. 4D) in Case 1 and Case 4. The injection sites were highly positive for protamine using IHC in Case 1 and Case 4, and the distribution of protamine was almost the same as insulin using double immunofluorescence staining (Fig. 5A, B and C). These results are listed in Table 6.

## 4. Discussion

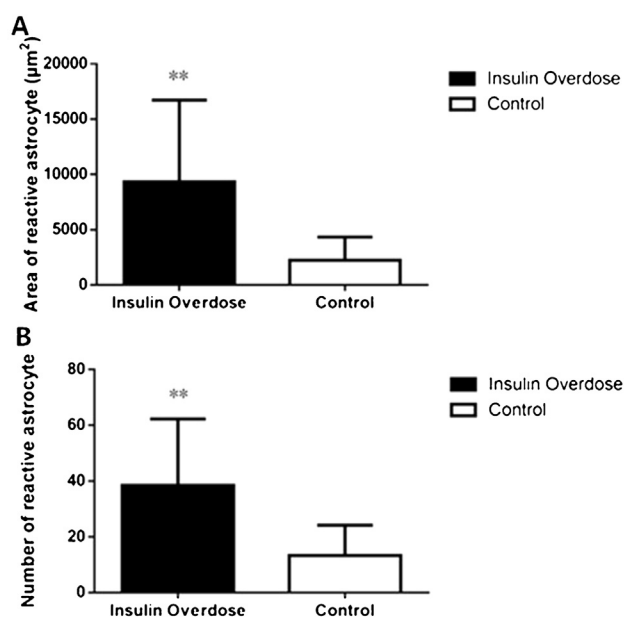
Toxicological screening is usually considered to be the gold standard in poisoning cases. However, postmortem analysis of insulin is not suitable for insulin dosage deduction in most cases because of the rapid occurrence of decomposition changes and blood hemolysis after death [9]. A recent study reveals that hemoglobin can break the disulfide bonds of insulin and lead to the new formation of intramolecular disulfide bridges, which reduces the concentration of insulin in postmortem blood samples [5]. In forensic practice, hemolysis is commonly encountered. As shown in Table 3, the blood insulin concentration of all four cases was lower than normal. Rapid separation of serum has been



**Fig. 2.** Reactive Astrocyte Proliferation in Cerebral Hemispheres. A: Reactive astrocyte proliferation in cerebral hemispheres, GFAP IHC-stained 1:200. B: Negative control, GFAP IHC-stained 1:200.

**Table 5**  
Reactive astrocytes proliferations in brain.

|                                                                   | Insulin overdose      | Control               |
|-------------------------------------------------------------------|-----------------------|-----------------------|
| Reactive astrocytes                                               | (+)-(++)              | (+)                   |
| Sum area of reactive astrocytes in each field ( $\mu\text{m}^2$ ) | $9348.60 \pm 7375.35$ | $2234.68 \pm 2102.69$ |
| Number of reactive astrocytes in each field                       | $38.4 \pm 23.84$      | $13.27 \pm 10.89$     |



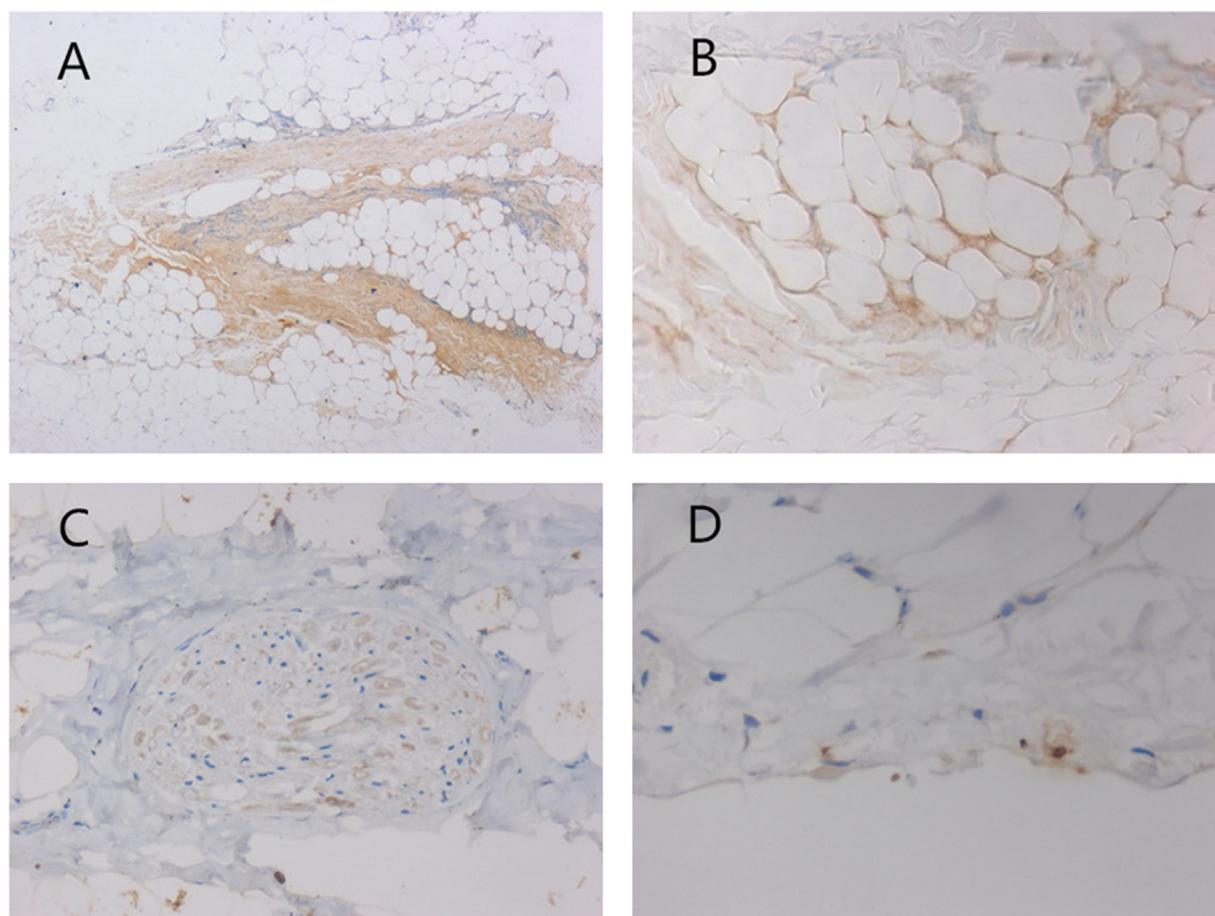
**Fig. 3.** Semi-quantification of the total area (3A) and the number (3B) of reactive astrocyte (\* $P < 0.01$ , \*\* $P < 0.001$ ).

demonstrated to be the best method to limit influence of hemolysis on postmortem insulin analysis [9]. Koskinen et al. uncovered an insulin overdose crime by detecting a high insulin concentration in stored frozen serum samples of the victims, emphasizing the

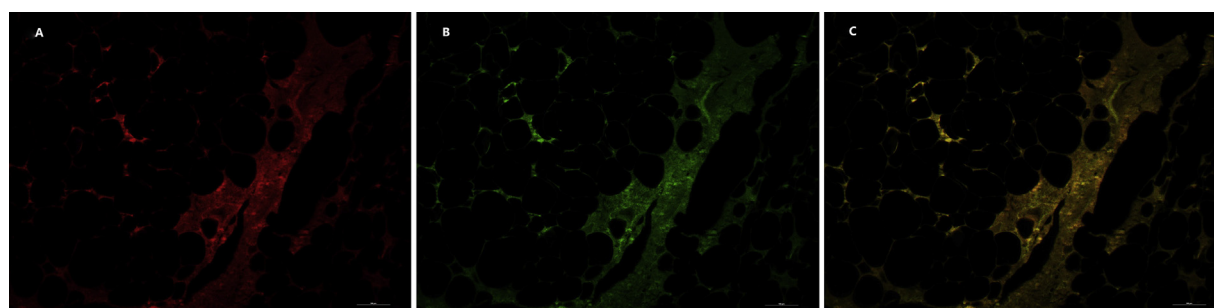
importance of storing serum samples appropriately in such cases [26]. Hypoglycemia is thought to be the first clue to insulin overdose in living subjects; however it is impossible to conclude that hypoglycemia was the cause of death based on postmortem glucose levels [27], because postmortem glucose concentrations are not reliable due to biochemical changes after death [6,8,28]. Table 3 lists all of the postmortem biochemical examinations of this study. Glucose concentrations in both blood and vitreous humor do not indicate hypoglycemia.

The mechanism of death due to insulin overdose is hypoglycemic encephalopathy which can give rise to irreversible CNS damage [6,26,29]. The clinical presentation of hypoglycemia consists of a CNS presentation of confusion, altered consciousness and memory disturbances as well as other symptoms such as sweating, rapid heartbeat and anxiety, which is described as acute neuroglycopenia [30,31]. Glucose is the major energy source for the functions of the brain. The normal glucose concentration of blood is 3.9–7.1 mmol/L, and when the blood glucose concentration is lower than 2.2 mmol/L, the glucose concentration in brain tissues is near zero.

Metabolic disorder is not the only pathogenesis of hypoglycemic brain damage. Apart from the decreased ATP level, accumulation of excitotoxic glutamate and aspartate damages the CNS as well [32,33]. Increased accumulation of aspartate, induced by a new neurochemical equilibrium in the brain, damages neurons [33], which is different from the mechanism of hypoxia-ischemia brain damage. Therefore, hypoglycemia can lead to a specific distribution of neuronal necrosis. Superficial laminae neurons are vulnerable to hypoglycemia while the middle laminae neurons are



**Fig. 4.** Distribution of insulin at injection sites. A: Insulin in the tissues of injection sites, Insulin IHC-stained 1:40. B: Insulin in gaps of adipocytes and connective tissues, Insulin-stained IHC 1:200. C: Insulin in peripheral nerves, Insulin IHC-stained 1:200. D: Insulin in inflammatory cells, Insulin IHC-stained 1:400.



**Fig. 5.** Double immunofluorescence staining of insulin with protamine. Distribution of insulin and protamine at injection sites, insulin and protamine IF-stained 1:40 (A: Insulin. B: Protamine. C: Merge).

**Table 6**

Distribution of insulin in the skin of injection sites.

| Case no. | Insulin detection in tissue homogenate | Distribution of insulin in skin tissue                                                                                                             | Protamine |
|----------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1        | /                                      | Gathering in gaps between adipocytes, in peripheral nerves, in dental connective tissue and on membrane of adipocytes                              | +         |
| 2        | +                                      | Gathering in gaps between adipocytes, in dental connective tissue and on membrane of adipocytes                                                    | –         |
| 3        | +                                      | /                                                                                                                                                  | /         |
| 4        | +                                      | Gathering in gaps between adipocytes, in peripheral nerves, in dental connective tissue, on membrane of adipocytes and in a few inflammatory cells | +         |

vulnerable to ischemia [34]. Auer et al. observed rarefied patches of neuropil devoid of neurons, focal GFAP positive astrocytes and Wallerian degeneration of axons with microglial nodules in real hypoglycemic coma cases [16]. Their controlled animal studies and clinical case report [35,36] also showed loss of granule cells in the dentate gyrus of hippocampus. They emphasized this pathologic change and recommended it as a marker for distinguishing hypoglycemia and hypoxia-ischemia. Kalimo and Olsson observed extensive necrosis in the cerebral cortex, basal ganglia, thalamus, caudal nucleus etc. and an increased number of astrocytes in regions of outer layers in two hypoglycemic coma cases [37]. In the brain stem and cerebellum, neuronal necrosis is generally minor under light microscopy [32]. Even at ultrastructural level, changes in Purkinje cells and granule cells of the cerebellum are minimal [38]. As all the victims in our study present acute insulin overdose and died within several hours, neuronal changes were not as typical as in the former research. We should emphasize that the astrogliosis is a forensic marker in insulin overdose. In our study, astrocytes rapidly reacted and proliferated after insulin administration. The change was observed in the cerebrum, corpus callosum, cerebellum and brain stem. Under conditions such as hypoglycemia, ischemia, brain trauma and encephalitis, astrocytes are activated to utilize more glutamate and provide glutamine and energy to maintain the function of neurons [32]. When this occurs, astrocytes become enlarged and the expression of GFAP increases [39]. The usefulness of GFAP IHC staining should be noted in the diagnosis of insulin overdose when ischemia, brain trauma and encephalitis are excluded.

Moderate edema and focal necrosis of hepatocytes demonstrated liver damage, which was conspicuous in Case 3 and Case 4. In the kidneys, many renal tubular epithelial cells became necrotic. Protein casts were found in some tubular cavities. In the lungs, focal edema and pulmonary hemorrhage were observed in many alveoli. These changes may be global presentations of the terminal stage, which are also found in other cases without insulin overdose. Many factors, such as the survival interval, body condition, age, etc., can affect such pathologic changes. Previous research shows that the myocardium is not sensitive to hypoglycemia and can function normally under such conditions [29]. This

may explain why there were no obvious pathologic changes in the heart.

In a review of 66 suspected, purported, or proven insulin homicide cases conducted by Marks, 66% of the alleged or suspected perpetrators worked in a hospital setting, such as nurses or physicians, and 78% were close relatives of the victims [6]. In our study, two of the four perpetrators were nurses and the other two perpetrators were an IDDM (Insulin-Dependent Diabetes Mellitus) patient and a daughter of an IDDM patient who knew the function of insulin well. In addition, all the perpetrators in this study had a close relationship with the victims and three perpetrators induced the victims to drink water with hypnotic pills before injecting the insulin. Finding empty insulin bottles at the primary scene was an indicator for insulin overdose. A previous study emphasizes that scene investigation is important in the evaluation of insulin-related death [40]. A coincidence is that all four perpetrators are female. It is possible that being weak in strength, female perpetrators most likely choose concealed methods in a crime.

The histopathologic analysis of suspected injection sites is a method to obtain proof of insulin administration [28,41]. Suspicious areas as well as control tissues without injection from the same corpse should be investigated. Birkinshaw et al. found injection tracks, hemorrhage, mild necrosis and inflammatory reactions [20]. Insulin IHC staining in skin tissue specimens is a suitable method to confirm the sites of insulin injection. According to previous clinical research, the effects of deposited insulin may last for several days, and the subcutaneous deposits can be surgically removed to treat hypoglycemia due to insulin injection [42–44]. In a fatal insulin overdose case report, Lutz et al. successfully detected insulin at the injection sites with a post-mortem interval of 24 days [28]. They found insulin-positive granular material surrounded by adipocyte membranes and interposed connective tissue [28]. Research also shows that insulin can be detected in inflammatory cells because of the phagocytosis of insulin by inflammatory cells at the injection sites [28,41]. Our IHC results are in accord with this previous research. In addition, we also found insulin in peripheral nerves, which has not been reported previously. The corpse in Case 1 was in water for 3 days



and had been frozen for 15 days before the autopsy, but we were still able to detect insulin at the injection site. In 1972, Phillips et al. applied a radioimmunoassay (RIA) technique to analyze insulin in the postmortem tissues of injection sites [45]. In the present study, homogenate using luminescence immunoassay, which further confirmed the morphologic results. Mudaliar et al. reported to have noticed insulin particles stained in the protamine during animal experiments [46], which led us to detect protamine as a substitute method. We observed protamine in the skin tissues of two victims, Case 1 and Case 4, because the injected insulin preparation contained protamine. The results of double immunofluorescence showed that the distribution of insulin and protamine was almost the same. This suggests that the detection of protamine can further confirm conclusions, and forensic pathologists should be aware of the dosage form of the insulin preparation in such cases.

## 5. Conclusion

Our study reveals that severe reactive astrocyte proliferation along with the presence of insulin at the injection sites can help diagnose insulin overdose. Insulin overdose can seriously damage the CNS due to acute hypoglycemia. At injection sites, insulin can be deposited in peripheral nerves in addition to the gaps between adipocytes, inflammatory cells and connective tissues. Forensic pathologists should be aware of the dosage form of the insulin preparation. We intend to emphasize that the detection of protamine at the injection sites is valuable in the diagnosis, if the injected insulin preparation contains protamine.

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## References

- [1] D.C. Winston, Suicide via insulin overdose in nondiabetics: the New Mexico experience, *Am. J. Forensic Med. Pathol.* 21 (3) (2000) 237–240.
- [2] W.C. Duckworth, Insulin degradation: mechanisms, products, and significance, *Endocr. Rev.* 9 (3) (1988) 319–345.
- [3] T. Morishima, S. Pye, C. Bradshaw, J. Radziuk, Posthepatic rate of appearance of insulin: measurement and validation in the nonsteady state, *Am. J. Physiol.* 263 (4 Pt 1) (1992) E772–E779.
- [4] D.R. Matthews, A.S. Rudenski, M.A. Burnett, P. Darling, R.C. Turner, The half-life of endogenous insulin and C-peptide in man assessed by somatostatin suppression, *Clin. Endocrinol.* 23 (1) (1985) 71–79.
- [5] C. Wunder, G.F. Kauert, S.W. Toennes, Factors leading to the degradation/loss of insulin in postmortem blood samples, *Forensic Sci. Int.* 241 (2014) 173–177.
- [6] V. Marks, Murder by insulin: suspected, purported and proven—a review, *Drug Test. Anal.* 1 (3–4) (2009) 162–176.
- [7] F. Musshoff, C. Hess, B. Madea, Disorders of glucose metabolism: post mortem analyses in forensic cases—part II, *Int. J. Leg. Med.* 125 (2) (2011) 171–180.
- [8] C. Palmiere, P. Mangin, Postmortem chemistry update part I, *Int. J. Leg. Med.* 126 (2) (2012) 187–198.
- [9] C. Palmiere, S. Sabatasso, C. Torrent, F. Rey, D. Werner, D. Bardy, Post-mortem determination of insulin using chemiluminescence enzyme immunoassay: preliminary results, *Drug Test. Anal.* 7 (9) (2015) 797–803.
- [10] T. Uezono, H. Shiono, K. Shimizu, K. Ogawa, O. Saito, M. Yoshida, H. Mizukami, K. Matsubara, Simultaneous analyses of hypoglycemic agents and C-peptide are essential in a homicide case with the combined dosing insulin and insulin-releasing drug, *Leg. Med. (Tokyo, Japan)* 4 (1) (2002) 34–36.
- [11] N. Sunderland, S. Wong, C.K. Lee, Fatal insulin overdoses: case report and update on testing methodology, *J. Forensic Sci.* 61 (Suppl 1) (2016) S281–S284.
- [12] V. Marks, G. Wark, Forensic aspects of insulin, *Diabetes Res. Clin. Pract.* 101 (3) (2013) 248–254.
- [13] H. Haibach, J.D. Dix, J.H. Shah, Homicide by insulin administration, *J. Forensic Sci.* 32 (1) (1987) 208–216.
- [14] I. Hood, B. Mirchandani, J. Monforte, W. Stacer, Immunohistochemical demonstration of homicidal insulin injection site, *Arch. Pathol. Lab. Med.* 110 (10) (1986) 973–974.
- [15] W.J. Levy, D. Gardner, J. Moseley, J. Dix, S.E. Gaede, Unusual problems for the physician in managing a hospital patient who received a malicious insulin overdose, *Neurosurgery* 17 (6) (1985) 992–996.
- [16] R.N. Auer, J. Hugh, E. Cosgrove, B. Curry, Neuropathologic findings in three cases of profound hypoglycemia, *Clin. Neuropathol.* 8 (2) (1989) 63–68.
- [17] S.D. Robinson, H. Safavi-Hemami, Insulin as a weapon, *Toxicol.* 123 (2016) 56–61.
- [18] I. Bourne, The insulin myth, *Lancet* 265 (6793) (1953) 964–968.
- [19] B. Ackner, A. Harris, A.J. Oldham, Insulin treatment of schizophrenia: a controlled study, *Lancet* 272 (6969) (1957) 607–611.
- [20] V.J. Birkinshaw, M.R. Gurd, S.S. Randall, A.S. Curry, D.E. Price, P.H. Wright, Investigations in a case of murder by insulin poisoning, *Br. Med. J.* 2 (5094) (1958) 463–468.
- [21] N.I. Batalis, J.A. Prahlow, Accidental insulin overdose, *J. Forensic Sci.* 49 (5) (2004) 1117–1120.
- [22] R. Jiang, L. Liu, W. Li, H. Liu, P. Wang, A case report of death due to insulin and estazolam, *J. Forensic Med.* 28 (04) (2012) 311.
- [23] H. Liu, C. Li, Smothering after insulin administration by vein: a case report, *J. Forensic Med.* 26 (04) (2010) 313.
- [24] K. Xu, M. Huang, Q. Lin, Murder by administration of large dose of insulin: a case analysis, *Forensic Sci. Technol.* 33 (04) (2008) 69–70.
- [25] Y. Yang, L. Zhang, J. Xing, W. Li, Y. Liang, F. Tong, Y. Duan, H. Dong, Y. Zhou, Homicidal insulin overdose: a case report, *Chin. J. Forensic Med.* 31 (02) (2016) 216–219.
- [26] P.J. Koskinen, H.M. Nuutinen, H. Laaksonen, J.A. Klossner, K.M. Irjala, H. Kalimo, J.S. Viikari, Importance of storing emergency serum samples for uncovering murder with insulin, *Forensic Sci. Int.* 105 (1) (1999) 61–66.
- [27] J.I. Coe, Postmortem chemistry update. Emphasis on forensic application, *Am. J. Forensic Med. Pathol.* 14 (2) (1993) 91–117.
- [28] R. Lutz, I. Pedal, C. Wetzel, R. Mattern, Insulin injection sites: morphology and immunohistochemistry, *Forensic Sci. Int.* 90 (1–2) (1997) 93–101.
- [29] W.F. Bobzien 3rd, Suicidal overdoses with hypoglycemic agents, *JACEP* 8 (11) (1979) 467–470.
- [30] V. Marks, Insulin and the law, *Ann. Clin. Biochem.* 52 (Pt 6) (2015) 696–698.
- [31] G. Beaumont, Automatism and hypoglycaemia, *J. Forensic Leg. Med.* 14 (2) (2007) 103–107.
- [32] S.W. Suh, A.M. Hamby, R.A. Swanson, Hypoglycemia, brain energetics, and hypoglycemic neuronal death, *Glia* 55 (12) (2007) 1280–1286.
- [33] R.N. Auer, Hypoglycemic brain damage, *Forensic Sci. Int.* 146 (2–3) (2004) 105–110.
- [34] R.N. Auer, B.K. Siesjö, Biological differences between ischemia, hypoglycemia, and epilepsy, *Ann. Neurol.* 24 (6) (1988) 699–707.
- [35] R.N. Auer, H. Kalimo, Y. Olsson, T. Wieloch, The dentate gyrus in hypoglycemia: pathology implicating excitotoxin-mediated neuronal necrosis, *Acta Neuropathol.* 67 (3–4) (1985) 279–288.
- [36] R.N. Auer, T. Wieloch, Y. Olsson, B.K. Siesjö, The distribution of hypoglycemic brain damage, *Acta Neuropathol.* 64 (3) (1984) 177–191.
- [37] H. Kalimo, Y. Olsson, Effects of severe hypoglycemia on the human brain. Neuropathological case reports, *Acta Neurol. Scand.* 62 (6) (1980) 345–356.
- [38] C.D. Agardh, H. Kalimo, Y. Olsson, B.K. Siesjö, Hypoglycemic brain injury: metabolic and structural findings in rat cerebellar cortex during profound insulin-induced hypoglycemia and in the recovery period following glucose administration, *J. Cereb. Blood Flow Metab.* 1 (1) (1981) 71–84.
- [39] K. Sahin, M. Tuzcu, C. Orhan, S. Ali, N. Sahin, H. Gencoglu, Y. Ozkan, A. Hayirli, N. Gozel, J.R. Komorowski, Chromium modulates expressions of neuronal plasticity markers and glial fibrillary acidic proteins in hypoglycemia-induced brain injury, *Life Sci.* 93 (25–26) (2013) 1039–1048.
- [40] M. Junge, M. Tsokos, K. Puschel, Suicide by insulin injection in combination with beta-blocker application, *Forensic Sci. Int.* 113 (1–3) (2000) 457–460.
- [41] F. Wehner, H.J. Mittmeyer, H.D. Wehner, M.C. Schieffer, Insulin- or morphine-injection? Immunohistochemical contribution to the elucidation of a case, *Forensic Sci. Int.* 95 (3) (1998) 241–246.
- [42] K. Chlud, On a suicide attempt with 2000 units of insulin novo lente, *Arch. Toxicol.* 21 (6) (1966) 333–337.
- [43] M.H. Samuels, R.H. Eckel, Massive insulin overdose: detailed studies of free insulin levels and glucose requirements, *J. Toxicol. Clin. Toxicol.* 27 (3) (1989) 157–168.
- [44] R. Guderjahn, Tissue excision at the injection site as a possible emergency therapy in suicide using depot-insulin, *Med. Welt* 27 (11) (1976) 538–539.
- [45] A.P. Phillips, B. Webb, A.S. Curry, The detection of insulin in postmortem tissues, *J. Forensic Sci.* 17 (3) (1972) 460–463.
- [46] S.R.D. Mudaliar, P. Mohideen, S.C. Baxi, M. Joyce, D.A. Armstrong, P. Strange, R. R. Henry, Pharmacodynamic and pharmacokinetic properties of a premixed 85/15 human insulin preparation, *Clin. Ther.* 23 (3) (2001) 404–412.